CareSwitch

Your movement. Your message.

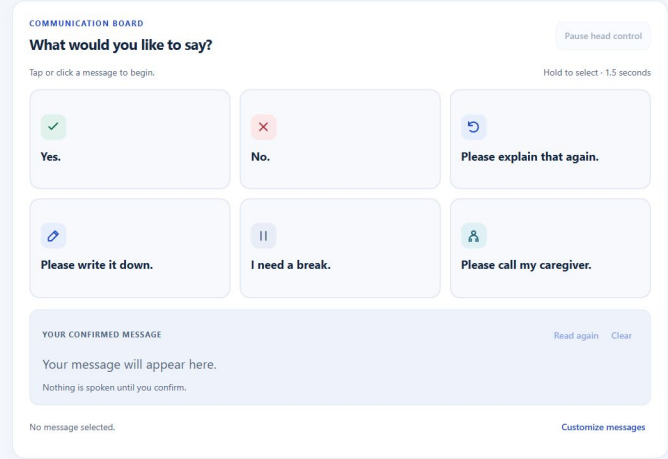
DSH Hacks V2 | AI x Healthcare | Solo student prototype

The problem and intended users

When speaking or using a touchscreen is difficult, expressing a simple preference can be hard. CareSwitch explores support for people who can voluntarily and comfortably move their head, with a helper for setup when needed. Intended-user testing is still required.

What the prototype does

An ordinary webcam controls an in-app pointer. Five-step calibration sets a comfortable range. Holding on one of six editable cards creates a draft; a separate Confirm or Cancel step keeps the choice explicit. Only confirmed messages are displayed and spoken.



Key features

Eight message packs: English, Urdu, Arabic, Hindi, Spanish, French, German and Chinese. Local device voices plus a free built-in speech fallback. Mouse, touch and keyboard alternatives. Responsive layout and adjustable dwell, sensitivity and smoothing.

Why this approach

A zero-paid-API prototype combining accessible input, explicit confirmation and multilingual messages in a browser. Existing assistive head-control tools inspired this work; no world-first claim is made.

AI and implementation

HTML/CSS/JavaScript, MediaPipe Face Landmarker and local Web Workers map a nose landmark to a filtered pointer. Dwell stops with missing, multiple or stale faces. Bundled eSpeak NG WebAssembly generates free speech locally; it is rule-based speech, not neural AI. AI tools assisted development and documentation.

Privacy, responsible boundaries and validation

Camera frames are not recorded or uploaded by the app; no patient database or remote speech service is used. This supports communication, not diagnosis, treatment or emergency alerts. The caregiver card speaks a request without contacting anyone. Automated and browser checks passed; the creator reported working laptop head control. Physical-phone use, pronunciation, fatigue and intended-user benefit remain unverified.

Next: feedback from intended users and accessibility specialists, fully translated setup instructions, and more input options.

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github.com/syedkaifali96/careswitch